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# EvalDetectBench: A Benchmark for Measuring Evaluation Awareness in Frontier Language Models

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## Abstract

Frontier large language models can often recognize when they are being evaluated, a capability known as evaluation awareness. If models behave differently in evaluations than in deployment, this undermines the validity of evaluation results, which are a crucial component of current AI safety frameworks. We introduce EvalDetectBench, an open pipeline and benchmark for measuring evaluation awareness that works with any Inspect-compatible evaluation, allowing practitioners to test against current and future benchmarks. EvalDetectBench ships with a newly curated transcript suite covering current frontier system-card evaluations and diverse deployment sources. The benchmark serves two purposes: measuring how reliably frontier LLMs recognize that they are being evaluated, and assessing how detectable individual benchmarks are as evaluations. We identify two methodological choices in the existing literature that introduce systematic bias: the identity of the model that generated the deployment transcripts accounts for 11.25% of measurement variance and can reorder model rankings; and elicitation prompts selected for high performance on one model can perform near chance on others. EvalDetectBench corrects for both via per-model probe calibration and a stratified generator-harmonisation procedure. We release EvalDetectBench as an open pipeline[3] and dataset[4].

## 1 Introduction

Evaluations inform deployment decisions, guide capability assessments, and provide the primary evidence base for safety cases across the industry [Anthropic, 2026c, Google DeepMind, 2025, OpenAI, 2025, xAI, 2025, Meta, 2026]. However, if a model can determine that it is being evaluated [Laine et al., 2024, Needham et al., 2025], it may alter its behaviour in ways that invalidate the measurement, performing differently from how it would in deployment environments.

This can cause under- or over-estimation of desired properties. In propensity evaluations, models may alignment-fake, behaving well because they are being tested [Greenblatt et al., 2024, Anthropic, 2025c], or behave badly because they detect that actions in a simulation will not have consequences.

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[3]https://github.com/freeze-lasr/aware_bench

[4]https://huggingface.co/datasets/el7982/aware-bench/

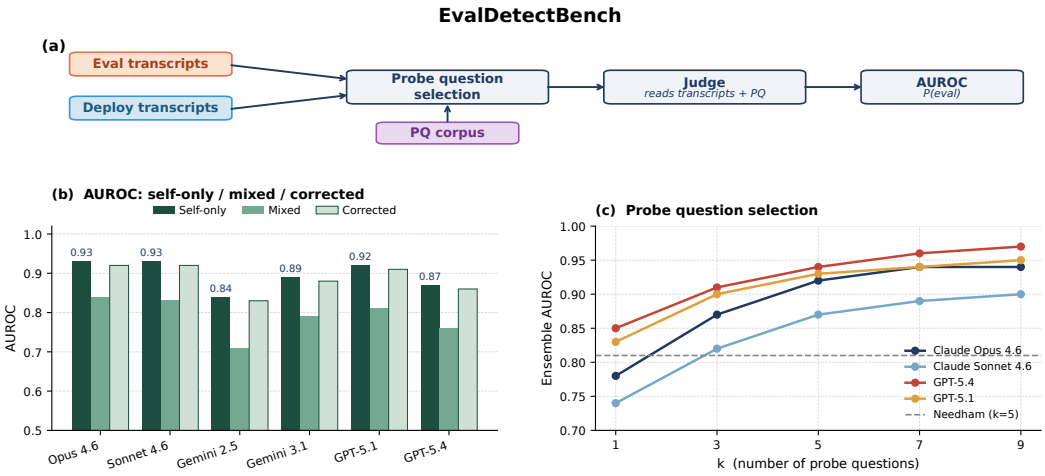



Figure 1: Overview of **EvalDetectBench** methodology. **(a)** Pipeline Architecture. **(b)** AUROC under three deployment-data conditions: **self-only** (unconfounded reference), **mixed** (mixture of various generator model deployment data), and **corrected**. **(c)** Per-model probe question selection.

In capability evaluations, models may sandbag, deliberately underperforming [van der Weij et al., 2025], or cheat by exploiting their knowledge that the task is an evaluation [Stein et al., 2026, Coleman, 2026]. In all cases, the evaluation no longer measures what it claims to.

Evaluation awareness has already been observed in deployed frontier models: Claude 3.7 Sonnet spontaneously reasons about being in alignment evaluations [Goldowsky-Dill et al., 2025], Sonnet 4.5 exhibits eval-gaming behaviour [Anthropic, 2025c], and metagaming reasoning emerges in o3 purely from capabilities RL [Schoen and Nitishinskaya, 2026]. The first systematic measurement of evaluation awareness as a capability [Needham et al., 2025] prompted a judge model to classify transcripts as evaluation or deployment, reporting the resulting $P(\text{eval})$. This established that models can discriminate at non-trivial rates, but left open whether the choice of prompts and the composition of the transcript pool generalise across model families and evaluation settings.

We introduce **EvalDetectBench** (Figure 1), a configurable pipeline with updated methodology and tooling to measure evaluation awareness capability in current frontier systems. The pipeline collects evaluation and deployment transcripts, scores each with prompts calibrated for the judge model, and reports both per-benchmark detectability and aggregate discrimination. Its modular design makes it straightforward to add new models, swap in new evaluations, and run controlled investigations into measurement validity (Section 4).

**Contributions:**

(1) **An open, extensible pipeline** for measuring evaluation awareness on any Inspect-compatible benchmark, with a refreshed transcript suite covering current frontier system-card evaluations and diverse deployment pools (Section 3).
(2) **Per-model probe calibration.** Elicitation prompts do not transfer across model families; a fixed set under-elicits awareness by up to 0.08 AUROC. Per-model selection outperforms fixed and random baselines (Section 3.2).
(3) **A generator-identity confound and correction.** 11.25% of $P(\text{eval})$ variance is attributable to the deployment transcript's generator; we show this persists under resampling and propose a stratified harmonisation that recovers the unconfounded signal (Section 3.3).
(4) **An empirical study** across 12 judge models and 13 benchmarks: judges discriminate at AUROC $> 0.75$ and $P(\text{eval})$ is high across benchmark categories (Section 4).

## 2 Background and Related Work

We define evaluation awareness as the capability to distinguish evaluation from deployment contexts, and elicit it via prompting (rather than fine-tuning or other techniques) so as not to overestimate.

Separately, models may have a *propensity* to consider whether they are being evaluated and act on that belief; we do not address propensity here, as it is a nascent measurement problem while capability evaluation is more established. Evaluation awareness was first noted anecdotally in Claude 3 Opus, which speculated mid-task that a retrieval prompt had been "inserted as a joke or to test if I was paying attention" [Anthropic, 2024], and has since been documented across recent frontier system cards.

**Chain-of-thought monitoring** detects *verbalised* awareness directly: Apollo Research observed Claude 3.7 Sonnet reasoning about evaluations in up to one-third of scheming runs [Goldowsky-Dill et al., 2025], and Anthropic's Sonnet 4.5 system card reports verbalised awareness rising from 20% to 80% across training [Anthropic, 2025c]. Verbalisation is a lower bound, however: non-reasoning models verbalise at under 1% without obviously behaving as though unaware [Goldowsky-Dill et al., 2025], and faithfulness limitations mean models may recognise an evaluation without verbalising [2026a, §4.5.5.3]. We focus on measuring evaluation awareness as a capability through prompting, to avoid limitations of chain-of-thought faithfulness and propensity to verbalise.

**White-box or interpretability methods** extract evaluation awareness signals from model activations: Nguyen et al. [2025] showed evaluation awareness is recoverable from Llama-3.3-70B-Instruct residual streams via linear probes, Chaudhary et al. [2025] found this scales predictably across four open-weight families, and Abdelnabi and Salem [2025], Hua et al. [2025], and Anthropic [2025c, 2026b, §6.5.2.2] further demonstrated that steering or ablating awareness-related directions causally shifts safety-relevant behaviour. We design our benchmark to be applicable to any model with API access, so it does not require white-box access of any kind, improving its applicability and reproducibility.

Closest to our work, **prompting-based** approaches directly ask a judge model to classify a transcript, which rests on the assumptions that the probe does not prime the model and that models respond honestly (discussed in Section 5). Prompting-based methods have been used in pre-deployment alignment evaluations[Kirk et al., 2026, Souly et al., 2026]. Laine et al. [2024] introduced the Situational Awareness Dataset covering seven components of situational awareness, one of these (the STAGES category) tests evaluation-vs-deployment classification on short text excerpts. Needham et al. focused on this evaluation awareness component specifically and built a dedicated benchmark of 1,000 transcripts from 61 sources, reporting AUROC up to 0.83 overall and 0.95 in agentic settings across nine frontier models. Ivanov [2025] compares black-box probing strategies on this dataset and finds disagreement between methods, motivating further methodological study. EvalDetectBench inherits Needham et al.'s prompted-classification framing, but revisits two design choices: the assumption that a single probe-question set transfers across models (Section 3.2) and the assumption that the transcript-generator's identity does not affect judge scores (Section 3.3.1), both of which we show meaningfully affect measurement. We also extend the Needham transcript suite with current frontier system-card benchmarks and a more diverse selection of deployment data (Section 3.1).

## 3 Methodology

We measure evaluation awareness by presenting a judge model with a transcript followed by a probe question (PQ) to elicit a classification of evaluation or deployment. We repeat over many transcripts (from both evaluation and deployment) and probe questions, aggregating the classifications. The pipeline has three components: a transcript suite of evaluation and deployment data (Section 3.1), a per-model probe-question selection procedure (Section 3.2), and scoring that combines probe responses into $P(\text{eval})$ and other metrics.

### 3.1 Data Collection

We diverge from Needham et al.'s dataset along three lines.

First, we move away from static transcripts: evaluation trajectories are collected fresh per evaluated model through our pipeline, so they reflect that model's own behaviour rather than a frozen log from a different model. This is important both for the ecological validity of the measurement, and also to avoid generator bias (Section 3.3.1).

Second, to support per-category analysis on current benchmarks, we add evaluations used in recent frontier model system cards to the set used to calibrate and measure performance.